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United States Patent

12408238

Kind Code

B2

Date of Patent

September 02, 2025

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Non-nicotine e-vaping section with channel and air passage traversing through first surface

Abstract

The non-nicotine e-vaping section includes a heater in heating proximity to a wick, the heater and the wick being connected to a first surface of a first wall that partially defines a chamber, and a reservoir configured to contain a non-nicotine pre-vapor formulation, the non-nicotine pre-vapor formulation being devoid of nicotine and including a non-nicotine compound, the first wall at least partially defining at least one first channel that traverses through the first surface, the at least one first channel being configured to communicate the non-nicotine pre-vapor formulation from the reservoir to the wick, the first wall defining an entrenched section that at least partially contains the wick, the first wall at least partially defining one or more air passages that traverse through the first surface, the one or more air passages having a first length that is longer than a second length of the at least one first channel.

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Appl. No.: 18/408165

Filed: January 09, 2024

Prior Publication Data

Document Identifier

Publication Date

US 20240163976 A1

May. 16, 2024

Related U.S. Application Data

Publication Classification

Int. Cl.: H05B3/03 (20060101); H05B1/02 (20060101)

U.S. Cl.:

CPC H05B3/03 (20130101); H05B1/0297 (20130101);

Field of Classification Search

CPC: A24F (40/42); A24F (40/485); H05B (3/03); H05B (1/0297)

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION (1) This application is a divisional of U.S. application Ser. No. 17/878,385, filed Aug. 1, 2022, which is a divisional of U.S. application Ser. No. 16/540,433, filed Aug. 14, 2019, the entire contents of each of which are incorporated herein by reference.

BACKGROUND

Field

(1) Example embodiments generally relate to a non-nicotine electronic vaping (e-vaping) section, and a non-nicotine e-vaping device including the non-nicotine e-vaping section.

Related Art

(2) A non-nicotine e-vaping device uses a heater to at least partially volatilize a non-nicotine pre-vapor formulation to produce a non-nicotine vapor.

SUMMARY

(3) At least one example embodiment is directed toward a non-nicotine e-vaping section.

(4) In one example embodiment, the non-nicotine e-vaping section include a housing; a wick in a chamber defined within the housing; a heater in heating proximity to the wick; and a reservoir configured to contain a non-nicotine pre-vapor formulation, the non-nicotine pre-vapor formulation being devoid of nicotine and including at least one non-nicotine compound, the non-nicotine e-vaping section defining at least one first channel, the at least one first channel being configured to communicate the non-nicotine pre-vapor formulation from the reservoir to the wick, and the non-nicotine e-vaping section further defining at least one first air passage, the at least one first air passage being configured to allow air to enter the reservoir.

(5) In one example embodiment, a total cross-sectional flow area of the at least one first channel is larger than a total cross-sectional flow area of the at least one first air passage.

(6) In one example embodiment, a total cross-sectional flow area of the at least one first channel is about 0.75 mm.² to 1.25 mm.² and a total cross-sectional flow area of the at least one first air passage is about 0.1 mm.² to 0.2 mm.².

(7) In one example embodiment, a ratio of a total cross-sectional flow area of the at least one first channel to a total cross-sectional flow area of the at least one first air passage is between about 9:1 and 5:1.

(8) In one example embodiment, a cross-sectional flow area of each one of the at least one first air passage is no larger than about 0.12 mm.².

(9) In one example embodiment, the wick does not extend into the reservoir and the wick does not extend into the at least one first channel.

(10) In one example embodiment, the at least one first channel includes two or more channels.

(11) In one example embodiment, at least one first air vent is defined within the non-nicotine e-vaping section, the at least one first air vent being configured to allow an airflow to enter the chamber.

(12) In one example embodiment, a discharge end of the at least one first air vent is positioned to directly face the heater.

(13) In one example embodiment, the at least one first air vent is configured to allow the airflow to enter the chamber in a first direction, and the chamber is configured to cause the airflow to flow at least partially across and away from the heater in a second direction, the first direction and the second direction being about perpendicular to each other.

(14) In one example embodiment, the heater includes at least one first flat heating surface, and the

first direction is about perpendicular to the at least one first flat heating surface.

(15) In one example embodiment, at least one first air inlet is defined by the housing, the at least one first air inlet being in fluid communication with the at least one first air vent if the non-nicotine e-vaping section is connected to a power section to form an non-nicotine e-vaping device.

(16) In one example embodiment, a first wall of the reservoir at least partially defines the at least one first channel and the at least one first air passage, and the wick is connected to an outer surface of the first wall, the wick covering a discharge end of the at least one first channel, the at least one first air passage including an inlet end that is positioned adjacent to the wick.

(17) In one example embodiment, the wick is connected to a wall of the chamber, the heater overlays and directly contacts the wick, and the heater includes at least one first flat heating surface that faces an interior of the chamber, the at least one first flat heating surface including openings that expose surface regions of the wick to the interior of the chamber.

(18) In one example embodiment, the wick is a thin pad.

(19) In one example embodiment, the non-nicotine e-vaping section further includes the non-nicotine pre-vapor formulation in the reservoir, wherein the non-nicotine pre-vapor formulation includes a non-nicotine vapor former, and the at least one non-nicotine compound.

(20) In one example embodiment, the at least one non-nicotine compound is *cannabis*, at least one *cannabis*-derived constituent, or both *cannabis* and the at least one *cannabis*-derived constituent.

(21) One example embodiment is directed toward a non-nicotine e-vaping device.

(22) In one example embodiment, the non-nicotine e-vaping section includes a non-nicotine e-vaping section, including, a housing, a wick in a chamber defined within the housing, a heater in heating proximity to the wick, and a reservoir configured to contain a non-nicotine pre-vapor formulation, the non-nicotine pre-vapor formulation being devoid of nicotine and including at least one non-nicotine compound, the non-nicotine e-vaping section defining at least one first channel, the at least one first channel being configured to communicate the non-nicotine pre-vapor formulation from the reservoir to the wick, and the non-nicotine e-vaping section further defining at least one first air passage, the at least one first air passage being configured to allow air to enter the reservoir; and a power section configured to connect to the non-nicotine e-vaping section, the power section including, a power source, and control circuitry, the control circuitry being configured to selectively send an electrical current from the power source to the heater.

(23) In one example embodiment, a total cross-sectional flow area of the at least one first channel is larger than a total cross-sectional flow area of the at least one first air passage.

(24) In one example embodiment, a total cross-sectional flow area of the at least one first channel is about 0.75 mm.² to 1.25 mm.² and a total cross-sectional flow area of the at least one first air passage is about 0.1 mm.² to 0.2 mm.².

(25) In one example embodiment, a ratio of a total cross-sectional flow area of the at least one first channel to a total cross-sectional flow area of the at least one first air passage is between about 9:1 and 5:1.

(26) In one example embodiment, a cross-sectional flow area of each one of the at least one first air passage is no larger than about 0.12 mm.².

(27) In one example embodiment, the wick does not extend into the reservoir and the wick does not extend into the at least one first channel.

(28) In one example embodiment, the at least one first channel includes two or more channels.

(29) In one example embodiment, at least one first air vent is defined within the non-nicotine e-vaping section, the at least one first air vent being configured to allow an airflow to enter the chamber, a discharge end of the at least one first air vent being positioned to directly face the heater.

(30) In one example embodiment, the at least one first air vent is configured to allow the airflow to enter the chamber in a first direction, and the chamber is configured to cause the airflow to flow at least partially across and away from the heater in a second direction, the first direction and the

second direction being about perpendicular to each other.

(31) In one example embodiment, at least one first air inlet is defined by the housing, the at least one first air inlet being in fluid communication with the at least one first air vent if the non-nicotine e-vaping section is connected to a power section to form an non-nicotine e-vaping device.

(32) In one example embodiment, a first wall of the reservoir at least partially defines the at least one first channel and the at least one first air passage, and the wick is connected to an outer surface of the first wall, the wick covering a discharge end of the at least one first channel, the at least one first air passage including an inlet end that is positioned adjacent to the wick.

(33) In one example embodiment, the wick is connected to a wall of the chamber, the heater overlays and directly contacts the wick, and the heater includes at least one first flat heating surface that faces an interior of the chamber, the at least one first flat heating surface including openings that expose surface regions of the wick to the interior of the chamber.

(34) In one example embodiment, the wick is a thin pad.

(35) In one example embodiment, the non-nicotine e-vaping device further includes a first pair of electrical connections on a first end of the non-nicotine e-vaping section; and a second pair of electrical connections on a second end of the power section, the first pair of electrical connections being mateable with the second pair of electrical connections to electrically connect the power source to the heater.

(36) In one example embodiment, the non-nicotine e-vaping device further includes at least one first sensor in the power section, the power section being in fluid communication with the chamber, the at least one first sensor being configured to measure at least one of a pressure drop, an airflow direction or both the pressure drop and the airflow direction; and circuitry, the circuitry being operationally connected to the at least one first sensor and the power source, the circuitry being configured to cause the power source to send the electrical current to the heater if the at least one first sensor senses a vaping condition.

(37) In one example embodiment, the non-nicotine e-vaping device further includes the non-nicotine pre-vapor formulation in the reservoir, wherein the non-nicotine pre-vapor formulation includes a non-nicotine vapor former, and the at least one non-nicotine compound.

(38) In one example embodiment, the at least one non-nicotine compound is *cannabis*, at least one *cannabis*-derived constituent, or both *cannabis* and the at least one *cannabis*-derived constituent.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) The various features and advantages of the non-limiting embodiments herein may become more apparent upon review of the detailed description in conjunction with the accompanying drawings. The accompanying drawings are merely provided for illustrative purposes and should not be interpreted to limit the scope of the claims. The accompanying drawings are not to be considered as drawn to scale unless explicitly noted. For purposes of clarity, various dimensions of the drawings may have been exaggerated.

(2) FIG. 1 is an illustration of a perspective view of an non-nicotine e-vaping device, in accordance with an example embodiment;

(3) FIG. 2 is an illustration of another perspective view of the non-nicotine e-vaping device, in accordance with an example embodiment;

(4) FIG. 3 is an illustration of an end of a first non-nicotine e-vaping section for the non-nicotine e-vaping device, in accordance with an example embodiment;

(5) FIG. 4 is an illustration of a perspective view of a power section for the non-nicotine e-vaping device, in accordance with an example embodiment;

(6) FIG. 5 is an illustration of a cut-away view of the non-nicotine e-vaping device, in accordance

with an example embodiment;

(7) FIG. 6 is an illustration of a cut-away view of the first non-nicotine e-vaping section, in accordance with an example embodiment;

(8) FIG. 7 is an illustration of another cut-away view of the first non-nicotine e-vaping section, in accordance with an example embodiment; and

(9) FIG. 8 is an illustration of another cut-away view of the first non-nicotine e-vaping section, in accordance with an example embodiment.

DETAILED DESCRIPTION OF EXAMPLE EMBODIMENTS

(10) Some detailed example embodiments are disclosed herein. However, specific structural and functional details disclosed herein are merely representative for purposes of describing example embodiments. Example embodiments may, however, be embodied in many alternate forms and should not be construed as limited to only the example embodiments set forth herein.

(11) Accordingly, while example embodiments are capable of various modifications and alternative forms, example embodiments thereof are shown by way of example in the drawings and will herein be described in detail. It should be understood, however, that there is no intent to limit example embodiments to the particular forms disclosed, but to the contrary, example embodiments are to cover all modifications, equivalents, and alternatives thereof. Like numbers refer to like elements throughout the description of the figures.

(12) It should be understood that when an element or layer is referred to as being “on,” “connected to,” “coupled to,” or “covering” another element or layer, it may be directly on, connected to, coupled to, or covering the other element or layer or intervening elements or layers may be present. In contrast, when an element is referred to as being “directly on,” “directly connected to,” or “directly coupled to” another element or layer, there are no intervening elements or layers present. Like numbers refer to like elements throughout the specification. As used herein, the term “and/or” includes any and all combinations or sub-combinations of one or more of the associated listed items.

(13) It should be understood that, although the terms first, second, third, etc. may be used herein to describe various elements, regions, layers and/or sections, these elements, regions, layers, and/or sections should not be limited by these terms. These terms are only used to distinguish one element, region, layer, or section from another region, layer, or section. Thus, a first element, region, layer, or section discussed below could be termed a second element, region, layer, or section without departing from the teachings of example embodiments.

(14) Spatially relative terms (e.g., “beneath,” “below,” “lower,” “above,” “upper,” and the like) may be used herein for ease of description to describe one element or feature's relationship to another element(s) or feature(s) as illustrated in the figures. It should be understood that the spatially relative terms are intended to encompass different orientations of the device in use or operation in addition to the orientation depicted in the figures. For example, if the device in the figures is turned over, elements described as “below” or “beneath” other elements or features would then be oriented “above” the other elements or features. Thus, the term “below” may encompass both an orientation of above and below. The device may be otherwise oriented (rotated 90 degrees or at other orientations) and the spatially relative descriptors used herein interpreted accordingly.

(15) The terminology used herein is for the purpose of describing various example embodiments only and is not intended to be limiting of example embodiments. As used herein, the singular forms “a,” “an,” and “the” are intended to include the plural forms as well, unless the context clearly indicates otherwise. It will be further understood that the terms “includes,” “including,” “comprises,” and/or “comprising,” when used in this specification, specify the presence of stated features, integers, steps, operations, and/or elements, but do not preclude the presence or addition of one or more other features, integers, steps, operations, elements, and/or groups thereof.

(16) When the words “about” and “substantially” are used in this specification in connection with a

numerical value, it is intended that the associated numerical value include a tolerance of $\pm 10\%$ around the stated numerical value, unless otherwise explicitly defined.

(17) Unless otherwise defined, all terms (including technical and scientific terms) used herein have the same meaning as commonly understood by one of ordinary skill in the art to which example embodiments belong. It will be further understood that terms, including those defined in commonly used dictionaries, should be interpreted as having a meaning that is consistent with their meaning in the context of the relevant art and will not be interpreted in an idealized or overly formal sense unless expressly so defined herein.

(18) Hardware may be implemented using processing or control circuitry such as, but not limited to, one or more processors, one or more Central Processing Units (CPUs), one or more microcontrollers, one or more arithmetic logic units (ALUs), one or more digital signal processors (DSPs), one or more microcomputers, one or more field programmable gate arrays (FPGAs), one or more System-on-Chips (SoCs), one or more programmable logic units (PLUs), one or more microprocessors, one or more Application Specific Integrated Circuits (ASICs), or any other device or devices capable of responding to and executing instructions in a defined manner.

(19) FIG. 1 is an illustration of a perspective view of an electronic vaping (e-vaping) device **10**, in accordance with an example embodiment. In an example embodiment, the non-nicotine e-vaping device **10** includes two sections: a first non-nicotine e-vaping section, cartridge, or pod **12**, and a power section **14**. In an example embodiment, the first non-nicotine e-vaping section **12** is connectable to the power section **14**. In another example embodiment, the non-nicotine e-vaping device **10** is one singular device that does not include separately connectable sections. In another example embodiment, the non-nicotine e-vaping device **10** includes more than two sections.

(20) In an example embodiment, the first non-nicotine e-vaping section **12** defines one or more outlets **16** on an end of the first non-nicotine e-vaping section **12**. In an example embodiment, the power section **14** includes at least one air inlet **18** for the non-nicotine e-vaping device **10**. In an example embodiment, the power section **14** includes one or more indicator lights **20** that indicate a capacity of the non-nicotine e-vaping device **10**, the power section **14** and/or the first non-nicotine e-vaping section **12**, where the capacity can include a power level, a non-nicotine pre-vapor formulation level, etc., as described herein in more detail. In an example embodiment, the one or more indicator lights **20** are light-emitting diodes (LEDs). In an example embodiment, the one or more indicator lights **20** are filament lights, incandescent lights, or other suitable types of lights.

(21) FIG. 2 is an illustration of another perspective view of the non-nicotine e-vaping device **10**, in accordance with an example embodiment. In an example embodiment, the non-nicotine e-vaping device **10** includes a power connector **22**. In an example embodiment, the power connector **22** may, for example, be a USB connector, a micro-USB connector, or another connector that connects the non-nicotine e-vaping device **10** to an electric power source.

Example Sections, According to Some Example Embodiments

(22) FIG. 3 is an illustration of an end of the first non-nicotine e-vaping section (pod) **12** for the non-nicotine e-vaping device **10**, in accordance with an example embodiment. In an example embodiment, the first non-nicotine e-vaping section **12** includes a first housing **13**. In an example embodiment, the first non-nicotine e-vaping section includes a connector **24** that is configured to connect the first non-nicotine e-vaping section **12** to the power section **14**. In an example embodiment, the connector **24** includes connecting structure **26** that includes one or more ribs **26a1** (as shown in FIG. 3). The one or more ribs **26a1** create a friction fit with the power section **14**. In an example embodiment, the connecting structure **26** may include tabs, magnets, detents, a latch, a snap fitting, or other suitable structure that cause the connector **24** to connect the first non-nicotine e-vaping section **12** to the power section **14**. In an example embodiment, the connector **24** connects the first non-nicotine e-vaping section **12** to the power section **14** via a friction fit.

(23) In an example embodiment, the first housing **13** defines at least one air inlet **36**. In an example embodiment, the first housing **13** of the connector **24** defines the at least one air inlet **36** which

aligns with the least one air inlet **18** defined by the power section **14** (see at least FIG. **4**). In an example embodiment, the first non-nicotine e-vaping section **12** includes a first end surface **32**, where a distal end **34** of the first housing **13** extends beyond the first end surface **32**. In an example embodiment, the distal end **34** of the first housing **13**, that extends beyond the first end surface **32**, defines the at least one air inlet **36**. In an example embodiment, the at least one air inlet **36** includes one air inlet, two air inlets, or more than two air inlets. In an example embodiment, a size of the at least one air inlet **36** is adjusted to control a desired resistance-to-draw (RTD) for the first non-nicotine e-vaping section **12**. In an example embodiment, the first end surface **32** at least partially defines at least one air vent **30**. In an example embodiment, the first non-nicotine e-vaping section **12** includes electrical contacts (electrical connections) **28**. In an example embodiment, the electrical contacts **28** are on the first end surface **32**.

(24) FIG. **4** is an illustration of a perspective view of the power section **14** for the non-nicotine e-vaping device **10**, in accordance with an example embodiment. In an example embodiment, the power section includes a housing **15**. In an example embodiment, a distal end **40** of the housing **15** extends beyond a third end surface **42** of the power section **14**. In an example embodiment, the distal end **40** of the housing **15**, that extends beyond the third end surface **42** of the power section **14**, defines the at least one air inlet **18**. In an example embodiment, the at least one air inlet **18** includes a pair of air inlets.

(25) In an example embodiment, the power section **14** includes electrical contacts (electrical connections) **44** that are able to mate with the electrical contacts **28** of the first non-nicotine e-vaping section **12**, once the first non-nicotine e-vaping section **12** is connected to the power section **14**. In an example embodiment, the power section **14** includes at least one hole **46** defined by the third end surface **42**.

(26) FIG. **5** is an illustration of a cut-away view of the non-nicotine e-vaping device **10**, in accordance with an example embodiment. In an example embodiment, the first non-nicotine e-vaping section **12** includes a reservoir **62** configured to contain a non-nicotine pre-vapor formulation **21** (see FIG. **6**). In an example embodiment, and as explained in more detail herein, a wick **64** is configured to absorb the non-nicotine pre-vapor formulation **21** and transport the non-nicotine pre-vapor formulation **21** from the reservoir **62** to a heater **60**. The heater **60** at least partially vaporizes the non-nicotine pre-vapor formulation **21** to form a non-nicotine vapor within a chamber **72**. A non-nicotine vapor, a non-nicotine aerosol and a non-nicotine dispersion are used interchangeably and refer to the matter generated or output by the devices and/or elements of the devices disclosed, claimed and/or equivalents thereof, that is devoid of nicotine. In an embodiment, the non-nicotine vapor in the chamber **72** is drawn from the chamber **72** via an airflow that passes through the at least one air vent **30**, through the chamber **72**, and out the one or more outlets **16**, as described herein in more detail (see fluid flow discussion in FIG. **6**).

(27) In an example embodiment, when the first non-nicotine e-vaping section **12** is connected to the power section **14**, an interior space **29** is defined between the first non-nicotine e-vaping section **12** and the power section **14**. Specifically, the interior space **29** is at least partially defined by the first end surface **32** of the first non-nicotine e-vaping section **12** (see FIG. **3**), the third end surface **42** of the power section **14** (see FIG. **4**) and the distal ends **34/40** of the respective first non-nicotine e-vaping section **12** and power section **14**. In an example embodiment, ambient air from outside of the non-nicotine e-vaping device **10** enters the interior space **29** through the at least one air inlet **18** in the power section **14** and the at least one air inlet **36** in the first non-nicotine e-vaping section **12**. In an example embodiment, the at least one air inlet **18** and the at least one air inlet **36** are at least partially aligned once the first non-nicotine e-vaping section **12** is connected to the power section **14**. In an example embodiment, the interior space **29** is in fluid communication with the at least one air vent **30** and the chamber **72**, and in fluid communication with the interior **53** of the power section **14** via the at least one hole **46**.

(28) In an example embodiment, the power section **14** includes a power source **50**. The power

source **50** may include a battery. In an example embodiment, the battery is a Lithium-ion battery or one of its variants, for example a Lithium-ion polymer battery. In an example embodiment, the battery is a Nickel-metal hydride battery, a Nickel cadmium battery, a Lithium-manganese battery, a Lithium-cobalt battery, a fuel cell or a solar cell. Any other power sources or battery technology may be used.

(29) In an example embodiment, the power section **14** includes a control system **58**. In an example embodiment, the control system **58** includes a controller **54** that is operationally connected to the power source **50** and at least one sensor **52**. In an example embodiment, the controller **54** of the control system **58** performs calculations and controls an operation of elements of the non-nicotine e-vaping device **10**, as described herein. In an example embodiment, the control system **58** includes control circuitry **55** that allows the power source **50** to be recharged. In an example embodiment, the at least one sensor **52** includes a pressure sensor and/or a temperature sensor. The at least one sensor **52** can be located in the power section **14** and/or the first non-nicotine e-vaping section **12**. In an example embodiment, the at least one sensor **52** is located in an interior **53** of the power section **14**. In an example embodiment, the at least one hole **46** (FIG. 4) causes the interior space **29** to be in fluid communication with the interior **53** of the power section **14**. In an example embodiment, the at least one sensor **52** is operationally constructed to measure one or more of the following: a resistance of the heater **60**, a temperature of the heater **60** and/or a draw of airflow through the non-nicotine e-vaping device **10**. In an example embodiment, the control system **58** receives an input signal, or signals, from the at least one sensor **52**, and the control system **58** controls operations of the non-nicotine e-vaping device **10**, including supplying an electrical current from the power source **50** to the heater **60** to vaporize the non-nicotine pre-vapor formulation **21** based at least in part on the signal(s) from the at least one sensor **52**. In an example embodiment, the control system **58** selectively causes the power source **50** to send an electrical current from the power source **50** to the one or more indicator lights **20**. In an example embodiment, the control system **58** is operationally and electrically connected to the heater **60** via the electrical contacts **28/44** that allow the control system **58** to selectively send the electrical current to the heater **60**. In an example embodiment, the control system **58** is operationally and electrically connected to the power connector **22** to control a charging regimen of the power source **50**.

(30) In an example embodiment, an airflow through the non-nicotine e-vaping device **10** activates the non-nicotine e-vaping device **10**. The at least one sensor **52** may be configured to generate an output indicative of an airflow, a magnitude of an airflow, and/or a direction of an airflow, where the control system **58** may receive output from the at least one sensor **52** and determine if the following internal conditions exist: (1) a direction of the airflow indicates a draw of airflow through the non-nicotine e-vaping device **10** (versus blowing air through the non-nicotine e-vaping device **10**), and/or (2) a magnitude of the airflow exceeds a threshold value. In some example embodiments, only one condition may be sufficient to activate the heater **60**, while in other examples, two conditions or all conditions may have to be met before activating the heater **60**. If these internal conditions of the non-nicotine e-vaping device **10** are met, the control system **58** electrically connects the power source **50** to the heater **60**, thereby activating the heater **60**. In an example embodiment, the at least one sensor **52** generates a variable output signal that is in at least partial correlation with a magnitude of a pressure drop sensed by the at least one sensor **52**. In an example embodiment, the control system **58** sends a variable electrical current to the heater **60** based on the variable output signal from the at least one sensor **52**. In an example embodiment, the control system **58** allows the heater **60** temperature to be manually selected.

(31) In an example embodiment, the control system **58** calculates a capacity of the non-nicotine e-vaping device **10**. In an example embodiment, the control system **58** performs this calculation through at least some input from the at least one sensor **52**. In an example embodiment, the control system **58** receives signals from the at least one sensor **52** that are indicative of an airflow traveling

through the non-nicotine e-vaping device **10**. In an example embodiment, the control system **58** includes one or more lookup tables that include tabulated data or values. Based on the received signal or signals from the at least one sensor **52**, and based on the one or more lookup tables, the control system **58** can calculate one or more of: a number of draws through the non-nicotine e-vaping device **10** or through the first non-nicotine e-vaping section **12**, a temperature of the heater **60**, a resistance of the heater **60**, a total and/or a cumulative volume of airflow through the non-nicotine e-vaping device **10** and/or the first non-nicotine e-vaping section **12**, a duration of use of the first non-nicotine e-vaping section **12**, a depletion of the non-nicotine pre-vapor formulation **21** in the reservoir **62**, a remaining capacity of the non-nicotine pre-vapor formulation **21** in the reservoir **62**, a dryness of the wick **64**, etc. In an example embodiment, the control system **58** calculates a capacity of the power source **50**. In an example embodiment, the control system **58** performs this calculation through at least some input from the at least one sensor **52**, in conjunction with the data or values from the one or more lookup tables. In an example embodiment, the control system **58** receives signals from the at least one sensor **52** and/or the control circuitry **55** that are indicative of an electrical current level output that is being discharged from the power source **50**. In an example embodiment, the control system **58** selectively sends an electrical current from the power source **50** to the one or more indicator lights **20** to visually reflect the result of one or more capacity determinations that is performed by the control system **58**.

(32) In an example embodiment, the power section **14** is used until the energy in the power source **50** is depleted and/or lowered below a certain threshold. In an example embodiment, the power source **50** is rechargeable and reusable, and the control circuitry **55** in the control system **58** allows the power source **50** to be charged by an external power source that connects to the power connector **22**. In an example embodiment, the power section **14** is rechargeable via solar power, or via an induction charging station. In some example embodiments, the control circuitry **55** of the control system **58**, when charged, provides power for a desired (or alternatively, a determined) number of draws, until the energy in power source **50** is depleted, and/or until the energy in power source **50** is lowered below a certain threshold, after which the control circuitry **55** must be re-connected to an external charging device.

(33) In an example embodiment, the first non-nicotine e-vaping section **12** is disposable. In this embodiment, the first non-nicotine e-vaping section **12** may be disposed of following depletion of the non-nicotine pre-vapor formulation **21** in the reservoir **62**. In an example embodiment, the first non-nicotine e-vaping section **12** is not disposable. In an example embodiment, the non-nicotine e-vaping device **10** is a single section, where the structure of the power section **14** and the first e-vaping section **12** are included in the single section. In an example embodiment, the non-nicotine e-vaping device **10** includes more than two sections.

(34) FIG. **6** is an illustration of a cut-away view of the first non-nicotine e-vaping section **12**, in accordance with an example embodiment. Previously described reference numbers are, in general, not described here again, for the sake of brevity. In an example embodiment, the first housing **13** and an interior housing **33** enclose internal elements of the first non-nicotine e-vaping section **12**. In an example embodiment, the interior housing **33** defines the at least one air vent **30**.

(35) In an embodiment, the reservoir **62** is defined by a first reservoir housing (wall) **37** and a second reservoir housing (wall) **39**. In an example embodiment, the first reservoir housing **37** includes a distal end portion **37a** that slides into interior walls **39a** of the second reservoir housing **39** to join the first reservoir housing **37** with the second reservoir housing **39** via a friction fit connection. In an example embodiment, a distal-most end **45** of the second reservoir housing **39** contacts a ledge **47** of the first housing **13**, where a cut-out region **43** holds a gasket **41** to form a liquid-tight seal between the first reservoir housing **37** and the second reservoir housing **39**. In an example embodiment, the reservoir **62** is defined by one continuous wall and/or housing, or more than two walls and/or housings. In an example embodiment, a capacity of the reservoir **62** provides enough non-nicotine pre-vapor formulation **21** for the first non-nicotine e-vaping section **12** to

produce about 10 to 20 draws, prior to disposal of the first non-nicotine e-vaping section **12**. In an example embodiment, a capacity of the reservoir **62** provides enough non-nicotine pre-vapor formulation **21** for the first non-nicotine e-vaping section **12** to produce more than 20 draws, prior to disposal of the first non-nicotine e-vaping section **12**.

(36) In an example embodiment, the first non-nicotine e-vaping section **12** includes channels **65** between the reservoir **62** and the chamber **72**. In an example embodiment, the first reservoir housing **37** defines one or more of the channels **65**. In an example embodiment, the channels **65** include one or more first channels (first micro-channels) **66** that are defined to exist between the reservoir **62** and the wick **64**. In an example embodiment, the one or more first channels **66** include only one channel, or two channels, or more than two channels. In an example embodiment, the one or more first channels **66** allow the wick **64** to transport a flow **67** of the non-nicotine pre-vapor formulation **21** from the reservoir **62** to the wick **64**, due at least in part to a capillary force provided by the wick **64**. In an example embodiment, the one or more first channels **66** allow the wick **64** to transport the flow **67** of the non-nicotine pre-vapor formulation **21** from the reservoir **62** to the wick **64**, due at least in part to a capillary force provided by the small diameter of the one or more first channels **66**. In an example embodiment, the flow **67** of the non-nicotine pre-vapor formulation **21** is assisted, at least in part, by an airflow **69** entering the reservoir **62**, as described below.

(37) In an example embodiment, the one or more first channels **66** includes at least two channels, to mitigate the possibility that the one or more first channels **66** becomes partially or fully obstructed by a bubble that may hinder or block the flow **67** of the non-nicotine pre-vapor formulation **21** from traveling through the one or more first channels **66**. In an example embodiment, the wick **64** does not extend directly into the reservoir **62**. In an example embodiment, no capillary structure or wicking system exists between the reservoir **62** and the wick **64**, or within the one or more first channels **66**, and the lone mode of transport of the non-nicotine pre-vapor formulation **21** from the reservoir **62** to the wick **64** is via communication through the one or more first channels **66**.

(38) In an example embodiment, the channels **65** include one or more second channels (air passage) **68**. In an example embodiment, the one or more second channels (second micro-channels) **68** include only one channel, or two channels, or more than two channels. In an example embodiment, the one or more second channels **68** are defined between the reservoir **62** and the chamber **72**. In an example embodiment, the one or more first channels **68** are positioned to be adjacent to the wick **64**. In an example embodiment, the one or more second channels **68** circumvent the wick **64** and/or the heater **60**. In an example embodiment, the one or more second channels **68** allow the airflow **69** to travel from the chamber **72** to the reservoir **62**, as the non-nicotine pre-vapor formulation **21** is displaced from the reservoir **62**. In an example embodiment, the airflow **69** is facilitated and/or assisted by pressure caused in the chamber **72** due to an incoming airflow **31** and a passing flow of non-nicotine vapor **73** within the chamber **72**. In an example embodiment, the airflow **69** is facilitated and/or assisted by a displacement (vacuum) force, as the non-nicotine pre-vapor formulation **21** is displaced and depleted from the reservoir **62**. In an example embodiment, the one or more second channels **68** are defined between the reservoir **62** and the non-nicotine vapor channel **70**, or another portion of the first non-nicotine e-vaping section **12** other than the chamber **72**, or ambient air.

(39) In an example embodiment, a first total cross-sectional flow area of the one or more first channels **66** is larger than a second total cross-sectional flow area of the one or more second channels **68**. In an example embodiment, a ratio of the first total cross-sectional area of the one or more first channels **66** and the second total cross-sectional area of the of the one or more second channels **68** is between about 10:1 and 4:1, or between about 9:1 and 5:1, or about 7:1. In an example embodiment, the first total cross-sectional flow area of the one or more first channels **66** is between about 0.5 mm.sup.2 to 1.5 mm.sup.2, or between about 0.75 mm.sup.2 to 1.25 mm.sup.2, or about 1 mm.sup.2. In an example embodiment, the second total cross-sectional flow area of the

one or more second channels **68** is between about 0.075 mm.sup.2 to 0.225 mm.sup.2, or between about 0.1 mm.sup.2 to 0.2 mm.sup.2, or about 0.15 mm.sup.2. In an example embodiment, each of the one or more second channels **68** is small enough that the non-nicotine pre-vapor formulation **21** is not able to travel through the one or more second channels **68**. The size of each of the one or more second channels **68** is dependent on factors that include: a smoothness of each of the one or more second channels **68**, a material that defines the one or more second channels **68** (the first reservoir housing **37**), a surface tension of the non-nicotine pre-vapor formulation **21**, etc. In an example embodiment, assuming that the one or more second channels **68** includes two channels, a cross-sectional flow area of each of the channels **68** is no larger than about 0.12 mm.sup.2, or no larger than about 0.1 mm.sup.2, or no larger than about 0.075 mm.sup.2. Other ranges of values for the sizes of the one or more first channels **66** and the one or more second channels **68**, and the ratio of the total cross-sectional flow area of the one or more first channels **66** and the one or more second channels **68**, are contemplated.

(40) In an example embodiment, the wick **64** is on a wall **76** of the chamber **72**. In an example embodiment, the wall **76** is formed at least in part by the first reservoir housing **37**. In an example embodiment, the wick **64** is embedded in the wall **76**, by an entrenched section **78** of the wall **76**. In an example embodiment, the heater **60** is in heating proximity to the wick **64**, such that the heater **60** is close enough to the wick **64** to at least partially vaporize the non-nicotine pre-vapor formulation **21** absorbed by the wick **64**. That is to say, the heater **60** is close enough to the wick **64** that the heater **60** is able to at least partially vaporize the non-nicotine pre-vapor formulation **21** that is absorbed by the wick **64**.

(41) In an example embodiment, the wick **64** is a thin pad. In an example embodiment, the wick **64** is rectangular. In an example embodiment, the wick **64** is square, circular, or another shape. In an example embodiment, the wick **64** is sized to absorb enough of the non-nicotine pre-vapor formulation **21** to produce one draw from the first non-nicotine e-vaping section **12**. In an example embodiment, the wick **64** is made of a porous material and/or absorbent material that has a capacity to absorb the non-nicotine pre-vapor formulation **21**. In an example embodiment, the wick **64** is made of fibrous materials, filaments, including glass or ceramic filaments. In an example embodiment, the wick **64** does not extend into the reservoir **62**. In an example embodiment, the wick **64** does not extend into the one or more first channels **66**. In an example embodiment, the wick **64** hold about 5 mm.sup.3 to 15 mm.sup.3 of the non-nicotine pre-vapor formulation **21**, or about 7.5 mm.sup.3 to 12.5 mm.sup.3, or about 10 mm.sup.3. In an example embodiment, the heater **60** and the wick **64** volatilize the non-nicotine pre-vapor formulation **21** in about 0.2 seconds.

(42) In an example embodiment, the heater **60** is in direct contact with the wick **64**. In an example embodiment, the heater **60** is on a surface of the wick **64**. In an example embodiment, the heater **60** includes a flat surface **80** that spans across at least a portion of a surface of the wick, as described in more detail in FIG. **8**. In an example embodiment, the flat surface **80** of the heater **60** faces an interior of the chamber **72**. In an example embodiment, the heater **60** is on a first side of the wick **64** that opposes a second side of the wick **64**, where the second side of the wick **64** faces the one or more first channels **66**.

(43) In an example embodiment, the at least one air vent **30** includes an outlet (discharge end) **30a** that directs the incoming airflow **31** at the heater **60**. In an example embodiment, the outlet **30a** is in close proximity to the heater **60**. In an example embodiment, the outlet **30a** is a distance of about 1.0 mm to 2.0 mm from the heater **60**, or about 1.2 mm to 1.5 mm from the heater **60**, or about 1.3 mm from the heater **60**. In an example embodiment, the outlet **30a** faces the heater **60**. In an example embodiment, the outlet **30a** directs the incoming airflow **31** at a center position **71** of the heater **60** (see FIG. **8**). In an example embodiment, the flow of non-nicotine vapor **73** in the chamber **72** passes across at least a portion of the heater **60**. In an example embodiment, facing the outlet **30a** at the heater **60** causes turbulent airflow conditions at the heater **60**, so that an intimate

mixing of the incoming airflow **31** and the non-nicotine vapor from the heater **60** can occur. In an example embodiment, the incoming airflow **31** is also directed at, towards or near the one or more second channels **68**, to provide an air pressure to further cause the airflow **69** to travel from the chamber **72** to the reservoir **62**. In an example embodiment, the incoming airflow **31** enters the chamber **72** in a first direction, and the chamber **72** causes the incoming airflow **31** to flow across and away from the heater **60** in a second direction, the first direction and the second direction being about perpendicular to each other. In an example embodiment, the first direction is about perpendicular to the flat surface **80** (see FIG. **8**) of the heater **60**. In an example embodiment, the second direction is about parallel to the flat surface **80** of the heater **60**. In an example embodiment, a size of the cross-sectional flow area of the at least one air vent **30** is adjusted to control the desired RTD for the first non-nicotine e-vaping section **12**. In an example embodiment, an inlet **30b** of the at least one air vent **30** is not directly exposed to ambient air during operation of the non-nicotine e-vaping device **10**, as ambient air instead first passes through the at least one air inlet **18** and the at least one air inlet **36** prior to reaching the at least one air vent **30**.

(44) In an example embodiment, a non-nicotine vapor channel **70** is defined within the first non-nicotine e-vaping section **12**. In an example embodiment, the non-nicotine vapor channel is at least partially defined by the first housing **13**, the first reservoir housing **37** and the second reservoir housing **39**. In an example embodiment, the non-nicotine vapor channel **70** is in fluid communication with the chamber **72** and the one or more outlets **16**, and the non-nicotine vapor channel **70** directs the flow of non-nicotine vapor **73** from the chamber **72** to the one or more outlets **16**.

(45) In an example embodiment, posts **74** and electrical contacts **75** electrically connect the electrical contacts **28** to the heater **60**.

(46) General Fluid Flow Through the First Non-Nicotine Section and Device, According to Some Embodiments:

(47) In an example embodiment, airflow enters the non-nicotine e-vaping device **10** through the at least one air inlet **18** (FIG. **4**), passes through the at least one air inlet **36**, the interior space **29** (FIG. **5**), the at least one air vent **30** and into the chamber **72**. In the chamber **72**, the airflow **31** picks up at least partially volatilized vapor from the heater **60**, and the resulting non-nicotine vapor **73** travels from the heater **60** and through the a non-nicotine vapor channel **70** prior to leaving the non-nicotine e-vaping device **10** through the one or more outlets **16**. In an example embodiment, while the first non-nicotine e-vaping section **12** is in use, the flow **67** of the non-nicotine pre-vapor formulation travels from the reservoir **62** through the one or more first channels **66** to the wick **64** to become at least partially volatilized by the heater **60**, while an airflow **69** enters the reservoir **62** via the one or more second channels **68**. FIG. **7** is an illustration of another cut-away view (perspective A-A, of FIG. **6**) of the first non-nicotine e-vaping section **12**, in accordance with an example embodiment. Previously described reference numbers are, in general, not described here again, for the sake of brevity. In an example embodiment, a distal end portion **33a** of the interior housing **33** at least partially defines the chamber **72**. In an example embodiment, the interior housing **33** fits into the first housing **13**, where a first contacting surface **77** of the interior housing **33** contacts the second contacting surface **76** to ensure the interior housing is positioned correctly in the first housing **13**. In an example embodiment, the interior housing **33** is held within the first housing **13** via a friction fit.

(48) In an example embodiment, the outlet **30a** faces the heater **60** and wick **64**, and the outlet **30a** is substantially centered on a center position **71** of the heater **60** (as shown in FIG. **8**).

(49) In an example embodiment, the distal end portion **37a** of the first reservoir housing **37** has an oval shaped cross-section, as shown in FIG. **7**.

(50) In an example embodiment, the second reservoir housing **39** is adhesively connected to the interior of the first housing **13**, and the first reservoir housing **37** is adhesively connected to the second reservoir housing **39**, via an application of an adhesive at one or more surface locations

where the second reservoir housing **39** and the first housing **13** contact each other, and at one or more surface locations where the first reservoir housing **37** and the second reservoir housing **39** contact each other. In an example embodiment, the interior housing **33** is adhesively connected to the first reservoir housing and/or the first housing **13** using an application of an adhesive at surface contact locations. In an example embodiment, the adhesive (sealant) is a silicon-based adhesive, or another suitable sealant, that provides a liquid and air-tight seal. In an example embodiment, the first reservoir housing **37**, the second reservoir housing **39**, the interior housing **33** and the first housing **13** are held together via a friction (press) fit, where no adhesive is used to assemble the first e-vaping section **12**.

(51) FIG. **8** is an illustration of a cut-away view (perspective B-B, of FIG. **7**) of the first non-nicotine e-vaping section **12**, in accordance with an example embodiment. This view shows elements along the wall **76** of the chamber **72** in better detail. Previously described reference numbers are, in general, not described here again, for the sake of brevity. In an example embodiment, the heater **60** includes a heating element **61**. In an example embodiment, the heating element **61** is a flat metallic structure. In an example embodiment, the heating element **61** is a thin and/or wire structure. In an example embodiment, the heating element **61** is in the shape of a wave (e.g., sinusoidal wave) or an “S” shape. In an example embodiment, the heating element **61** is shaped to maximize a surface contact with the wick **64**. In an example embodiment, the heating element defines openings **82** within the heating element **61**, where the openings expose surface regions of the wick **64** to the interior of the chamber **72**. In an example embodiment, the heater **60**, or the heating element **61** of the heater **60**, forms a substantially flat surface **80**. In an example embodiment, the heater **60** is constructed of an iron-aluminide (e.g., FeAl or Fe.sub.3Al). In an embodiment, the heater **60** is in the form of a wire coil, a planar body, a ceramic body, a single wire, a cage of resistive wire, or any other suitable form that is configured to vaporize the non-nicotine pre-vapor formulation **21**. In at least one example embodiment, the heater **60** is formed of any suitable electrically resistive material or materials. In an example embodiment, the heater **60** is a ceramic heater having an electrically resistive layer on an outside surface thereof.

(52) In an example embodiment, the center position **71** of the heater **60**, which is a portion of the heater **60** that the outlet **30a** faces, includes the flat surface **80**. In an example embodiment, the center position **71** of the heater **60** corresponds to a center area of the heater **60** and/or the heating element **61**.

(53) In an example embodiment, the heater **60** includes electrical contacts **84**. In an example embodiment, the electrical contacts **84** are electrically connected to the power source **50**. In an example embodiment, the electrical contacts **84** of the heater **60** are electrically connected to the electrical contacts **75** and the posts **74** of the first non-nicotine e-vaping section **12**, where the posts **74** are in turn electrically connected to the electrical contacts **28** of the first non-nicotine e-vaping section **12** and the electrical contacts **44** of the power section **14**. In an example embodiment, one of the electrical contacts **44** is electrically connected to the power source **50** and the other electrical contact **44** is connected to the control circuitry **55**, so that the control circuitry **55** of the control system **58** can selectively cause the power source **50** to send an electrical current to the electrical contacts **44** of the power section, through the electrical contacts **28** of the first non-nicotine e-vaping section **12**, and through the posts **74** and the electrical contacts **75** to energize the heater **60**.

(54) In an example embodiment, the one or more second channels **68** are on sides of the wick **64**. In an example embodiment, the one or more second channels **68** are not covered by the wick **64**, and the one or more first channels **66** are covered by the wick **64**.

Advantages of Some of the Example Embodiments

(55) Advantageous of some of the example embodiments include the following.

(56) A. Gravity independence: Some factors including a relatively small non-nicotine pre-vapor formulation mass, a small size of the one or more first channels **66**, and a geometry of elements of the first non-nicotine e-vaping section **12**, at least partially assist in causing the non-nicotine e-

vaping device **10** to be less dependent, or not dependent, on gravity in order to operate the non-nicotine e-vaping device **10** and communicate the non-nicotine pre-vapor formulation **21** to the wick **64** and heater **60**. That is to say, an orientation of the non-nicotine e-vaping device **10** does not impact or change a performance of the non-nicotine e-vaping device **10**. These factors at least partially assist in mitigating leakage and ensuring that a desired and uniform amount of the non-nicotine pre-vapor formulation **21** is applied to the wick and vaporized by the heater **60**.

(57) B. Reduced power source size: Some factors including a relatively small non-nicotine pre-vapor formulation mass and a geometry of elements of the first non-nicotine e-vaping section **12** allow for a relatively small power source **50**. This can assist charging schemes for the non-nicotine e-vaping device **10**.

Example Embodiments with Non-Nicotine Pre-Vapor Formulation

(58) In an example embodiment, a flavoring (at least one flavorant) and/or a non-nicotine compound is included in the non-nicotine pre-vapor formulation **21**. In an example embodiment, the non-nicotine pre-vapor formulation **21** is a liquid, solid, dispersion and/or a gel formulation including, but not limited to, water, beads, solvents, active ingredients, ethanol, plant extracts, natural or artificial flavors, and/or at least one non-nicotine vapor former such as glycerin and propylene glycol.

(59) The non-nicotine compound is devoid of nicotine. In an example embodiment, the non-nicotine compound does not include tobacco, nor is the compound derived from tobacco. In an example embodiment, the non-nicotine compound is *cannabis*, or includes at least one *cannabis*-derived constituent. In an example embodiment, a *cannabis*-derived constituent includes at least one of a *cannabis*-derived cannabinoid (e.g., a phytocannabinoid, or a cannabinoid synthesized by a *cannabis* plant), at least one *cannabis*-derived terpene, at least one *cannabis*-derived flavonoid, or combinations thereof.

(60) In an example embodiment, the non-nicotine compound is in the form of, or included in, a solid, a semi-solid, a gel, a hydrogel, or combinations thereof, and the non-nicotine compound is infused into, or co-mingled or combined within, the non-nicotine pre-vapor formulation **21**. In an example embodiment, the non-nicotine compound is in the form of, or included in, a liquid or a partial-liquid, that includes an extract, an oil, a tincture, a suspension, a dispersion, a colloid, an alcohol, a general non-neutral (slightly acidic or slightly basic) solution, or combinations thereof, and the non-nicotine compound is infused into, or comingled or combined within, the non-nicotine pre-vapor formulation **21**. In an example embodiment, the non-nicotine compound is a constituent of the non-nicotine pre-vapor formulation **21**. In an example embodiment, the non-nicotine pre-vapor formulation **21** is, or is part of, a dispersion, a suspension, a gel, a hydrogel, a colloid, or combinations thereof, and the non-nicotine compound is a constituent of the non-nicotine pre-vapor formulation **21**.

(61) In an example embodiment, the non-nicotine compound undergoes a slow, natural decarboxylation process over an extended duration of time at low temperatures, including at or below room temperature (72° F.). In an example embodiment, the non-nicotine compound may undergo a significantly elevated decarboxylation process, on the order of 50% decarboxylation or greater if the non-nicotine compound is exposed to elevated temperatures especially in the range of about 175° F. or greater over a period of time (minutes or hours, at a relatively low pressure such as 1 atmosphere), where even further elevated temperatures (about 240° F. or greater) can cause a rapid or instantaneous decarboxylation to occur at a potentially high decarboxylation rate (50% or more), though even further elevated temperatures can cause a degradation of some or all of the chemical properties of the non-nicotine compounds.

(62) In an example embodiment, the at least one non-nicotine vapor former of the non-nicotine pre-vapor formulation includes diols (such as propylene glycol and/or 1,3-propanediol), glycerin and combinations, or sub-combinations, thereof. Various amounts of non-nicotine vapor former may be used. For example, in some example embodiments, the at least one non-nicotine vapor former is

included in an amount ranging from about 20% by weight based on the weight of the non-nicotine pre-vapor formulation **21** to about 90% by weight based on the weight of the non-nicotine pre-vapor formulation **21** (for example, the non-nicotine vapor former is in the range of about 50% to about 80%, or about 55% to 75%, or about 60% to 70%), etc. As another example, in an example embodiment, the non-nicotine pre-vapor formulation **21** includes a weight ratio of the diol to glycerin that ranges from about 1:4 to 4:1, where the diol is propylene glycol, or 1,3-propanediol, or combinations thereof. In an example embodiment, this ratio is about 3:2. Other amounts or ranges may be used.

(63) In an example embodiment, the non-nicotine pre-vapor formulation **21** includes water. Various amounts of water may be used. For example, in some example embodiments, water may be included in an amount ranging from about 5% by weight based on the weight of the non-nicotine pre-vapor formulation **21** to about 40% by weight based on the weight of the non-nicotine pre-vapor formulation **21**, or in an amount ranging from about 10% by weight based on the weight of the non-nicotine pre-vapor formulation **21** to about 15% by weight based on the weight of the non-nicotine pre-vapor formulation **21**. Other amounts or percentages may be used. For example, in an example embodiment, the remaining portion of the non-nicotine pre-vapor formulation **21** that is not water (and not the non-nicotine compound and/or flavorants), is the non-nicotine vapor former (described above), where the non-nicotine vapor former is between 30% by weight and 70% by weight propylene glycol, and the balance of the non-nicotine vapor former is glycerin. Other amounts or percentages may be used.

(64) In an example embodiment, the non-nicotine pre-vapor formulation **21** includes at least one flavorant in an amount ranging from about 0.2% to about 15% by weight (for instance, the flavorant may be in the range of about 1% to 12%, or about 2% to 10%, or about 5% to 8%). In an example embodiment, the at least one flavorant includes volatile *cannabis* flavor compounds (flavonoids). In an example embodiment, the at least one flavorant includes flavor compounds instead of, or in addition to, the *cannabis* flavor compounds. In an example embodiment, the at least one flavorant may be at least one of a natural flavorant, an artificial flavorant, or a combination of a natural flavorant and an artificial flavorant. For instance, the at least one flavorant may include menthol, wintergreen, peppermint, cinnamon, clove, combinations thereof, and/or extracts thereof. In addition, flavorants may be included to provide herb flavors, fruit flavors, nut flavors, liquor flavors, roasted flavors, minty flavors, savory flavors, combinations thereof, and any other desired flavors.

(65) In an example embodiment, the non-nicotine compound may be a medicinal plant, or a naturally occurring constituent of the plant that has a medically-accepted therapeutic effect. The medicinal plant may be a *cannabis* plant, and the constituent may be at least one *cannabis*-derived constituent. Cannabinoids (phytocannabinoids) are an example of a *cannabis*-derived constituent, and cannabinoids interact with receptors in the body to produce a wide range of effects. As a result, cannabinoids have been used for a variety of medicinal purposes. *Cannabis*-derived materials may include the leaf and/or flower material from one or more species of *cannabis* plants, or extracts from the one or more species of *cannabis* plants. In an example embodiment, the one or more species of *cannabis* plants includes *Cannabis sativa*, *Cannabis indica*, and *Cannabis ruderalis*. In some example embodiments, the non-nicotine pre-vapor formulation **21** includes a mixture of *cannabis* and/or *cannabis*-derived constituents that are, or are derived from, 60-80% (e.g., 70%) *Cannabis sativa* and 20-40% (e.g., 30%) *Cannabis indica*.

(66) Examples of *cannabis*-derived cannabinoids include tetrahydrocannabinolic acid (THCA), tetrahydrocannabinol (THC), cannabidiolic acid (CBDA), cannabidiol (CBD), cannabinol (CBN), cannabicyclol (CBL), cannabichromene (CBC), and cannabigerol (CBG). Tetrahydrocannabinolic acid (THCA) is a precursor of tetrahydrocannabinol (THC), while cannabidiolic acid (CBDA) is precursor of cannabidiol (CBD). Tetrahydrocannabinolic acid (THCA) and cannabidiolic acid (CBDA) may be converted to tetrahydrocannabinol (THC) and cannabidiol (CBD), respectively,

via heating. In an example embodiment, heat from the heater **60** may cause decarboxylation to convert tetrahydrocannabinolic acid (THCA) in the non-nicotine pre-vapor formulation **21** to tetrahydrocannabinol (THC), and/or to convert cannabidiolic acid (CBDA) in the non-nicotine pre-vapor formulation **21** to cannabidiol (CBD).

(67) In instances where both tetrahydrocannabinolic acid (THCA) and tetrahydrocannabinol (THC) are present in the non-nicotine pre-vapor formulation **21**, the decarboxylation and resulting conversion will cause a decrease in tetrahydrocannabinolic acid (THCA) and an increase in tetrahydrocannabinol (THC). At least 50% (e.g., at least 87%) of the tetrahydrocannabinolic acid (THCA) may be converted to tetrahydrocannabinol (THC), via the decarboxylation process, during the heating of the non-nicotine pre-vapor formulation **21** for purposes of vaporization. Similarly, in instances where both cannabidiolic acid (CBDA) and cannabidiol (CBD) are present in the non-nicotine pre-vapor formulation **21**, the decarboxylation and resulting conversion will cause a decrease in cannabidiolic acid (CBDA) and an increase in cannabidiol (CBD). At least 50% (e.g., at least 87%) of the cannabidiolic acid (CBDA) may be converted to cannabidiol (CBD), via the decarboxylation process, during the heating of the non-nicotine pre-vapor formulation **21** for purposes of vaporization.

(68) The non-nicotine pre-vapor formulation **21** may contain the non-nicotine compound that provides the medically-accepted therapeutic effect (e.g., treatment of pain, nausea, epilepsy, psychiatric disorders). Details on methods of treatment may be found in U.S. application Ser. No. 15/845,501, filed Dec. 18, 2017, titled “VAPORIZING DEVICES AND METHODS FOR DELIVERING A COMPOUND USING THE SAME,” the disclosure of which is incorporated herein in its entirety by reference.

(69) In an example embodiment, the portion of the non-nicotine pre-vapor formulation **21** that is not the non-nicotine compound and/or the flavorant, includes 10-15% by weight water, where the remaining portion of the non-nicotine compound and non-flavorant portion of the non-nicotine pre-vapor formulation **21** is a mixture of propylene glycol and a non-nicotine vapor former where the mixture is in a ratio that ranges between about 60:40 and 40:60 by weight. Other combinations, amounts or ranges may be used.

(70) Example embodiments have been disclosed herein, it should be understood that other variations may be possible. Such variations are not to be regarded as a departure from the spirit and scope of the present disclosure, and all such modifications as would be obvious to one skilled in the art are intended to be included within the scope of the following claims.

Claims

1. A non-nicotine e-vaping section, comprising: a heater in heating proximity to a wick, the heater and the wick being connected to a first surface of a first wall that partially defines a chamber; and a reservoir configured to contain a non-nicotine pre-vapor formulation, the non-nicotine pre-vapor formulation being devoid of nicotine and including at least one non-nicotine compound, the first wall at least partially defining at least one first channel that traverses through the first surface, the at least one first channel being configured to communicate the non-nicotine pre-vapor formulation from the reservoir to the wick, and the first wall defining an entrenched section that at least partially contains the wick, the first wall at least partially defining one or more air passages that traverse through the first surface, the one or more air passages having a first length that is longer than a second length of the at least one first channel.

2. The non-nicotine e-vaping section of claim 1, wherein the one or more air passages are configured to allow air to enter the reservoir, the at least one first channel and the one or more air passages being configured to respectively communicate the non-nicotine pre-vapor formulation and the air in opposing directions through the first wall, and the at least one first channel and the one or more air passages run parallel to each other.

3. The non-nicotine e-vaping section of claim 1, wherein a total cross-sectional flow area of the at least one first channel is larger than a total cross-sectional flow area of the one or more air passages.
4. The non-nicotine e-vaping section of claim 1, wherein the wick does not extend into the reservoir and the wick does not extend into the at least one first channel.
5. The non-nicotine e-vaping section of claim 1, wherein at least one first air vent is defined within the non-nicotine e-vaping section, the at least one first air vent being configured to allow an airflow to enter the chamber.
6. The non-nicotine e-vaping section of claim 5, wherein a discharge end of the at least one first air vent is positioned to directly face the heater.
7. The non-nicotine e-vaping section of claim 5, wherein the at least one first air vent is configured to allow the airflow to enter the chamber in a first direction, and the chamber is configured to cause the airflow to flow at least partially across and away from the heater in a second direction, the first direction and the second direction being about perpendicular to each other.
8. The non-nicotine e-vaping section of claim 7, wherein the heater includes at least one first flat heating surface, and the first direction is about perpendicular to the at least one first flat heating surface.
9. The non-nicotine e-vaping section of claim 5, wherein at least one first air inlet is in fluid communication with the at least one first air vent if the non-nicotine e-vaping section is connected to a power section to form a non-nicotine e-vaping device.
10. The non-nicotine e-vaping section of claim 1, wherein the wick covers a discharge end of the at least one first channel, the one or more air passages including an inlet end that is positioned adjacent to the wick.
11. The non-nicotine e-vaping section of claim 1, wherein the heater overlays and directly contacts the wick, and the heater includes at least one first flat heating surface that faces an interior of the chamber, the at least one first flat heating surface including openings that expose surface regions of the wick to the interior of the chamber.
12. The non-nicotine e-vaping section of claim 1, wherein the wick is a pad.
13. The non-nicotine e-vaping section of claim 1, further comprising: the non-nicotine pre-vapor formulation in the reservoir, wherein the non-nicotine pre-vapor formulation further includes a non-nicotine vapor former.
14. The non-nicotine e-vaping section of claim 1, wherein the one or more air passages are spaced apart from the at least one first channel, the wick not extending into the reservoir and the wick not extending into the at least one first channel, the one or more air passages being spaced apart from the wick.
15. The non-nicotine e-vaping section of claim 1, wherein an entrance of a first air passage, of the one or more air passages, is between the heater and an outlet of the non-nicotine e-vaping section, the heater being configured to at least partially vaporize the non-nicotine pre-vapor formulation that is communicated to the wick, the chamber, the heater and the entrance being configured to allow a portion of the vapor to enter the reservoir through the first air passage.
16. A non-nicotine e-vaping device, comprising: a non-nicotine e-vaping section, including, a heater in heating proximity to a wick, the heater and the wick being connected to a first surface of a first wall that partially defines a chamber; and a reservoir configured to contain a non-nicotine pre-vapor formulation, the non-nicotine pre-vapor formulation being devoid of nicotine and including at least one non-nicotine compound, the first wall at least partially defining at least one first channel that traverses through the first surface, the at least one first channel being configured to communicate the non-nicotine pre-vapor formulation from the reservoir to the wick, and the first wall defining an entrenched section that at least partially contains the wick, the first wall at least partially defining one or more air passages that traverse through the first surface, the one or more air passages having a first length that is longer than a second length of the at least one first channel;

and a power section configured to connect to the non-nicotine e-vaping section, the power section including, a power source, and control circuitry, the control circuitry being configured to selectively send an electrical current from the power source to the heater.

17. The non-nicotine e-vaping device of claim 16, wherein the one or more air passages are configured to allow air to enter the reservoir, the at least one first channel and the one or more air passages being configured to respectively communicate the non-nicotine pre-vapor formulation and the air in opposing directions through the first wall, and the at least one first channel and the one or more air passages run parallel to each other.

18. The non-nicotine e-vaping device of claim 16, wherein a total cross-sectional flow area of the at least one first channel is larger than a total cross-sectional flow area of the one or more air passages.

19. The non-nicotine e-vaping device of claim 16, wherein the wick does not extend into the reservoir and the wick does not extend into the at least one first channel.
